

RAHUL MITRA

CYBER SECURITY PROFILE

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PROFILE SUMMARY

Early-career cyber security engineer and bug bounty researcher with hands-on vulnerability discovery across Singapore government and transport programs. Combines adversarial testing, automation scripting, secure design awareness, and software engineering depth.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Software Engineering; Artificial Intelligence; 3D Computer Vision; Computer Systems; Statistics; Mathematics; Engineering Systems
- Certificate of Outstanding Performance in 3D Computer Vision (CS4277) - top student in a class of 24, NUS School of Computing.

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERNSHIP EXPERIENCE

Operational AI Systems & Data Engineer (Contract), Abbott Laboratories Singapore Pte Ltd

Jun 2026 - Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.

Data Analytics and Project/Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Jun 2026

- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.

Independent Bug Bounty Researcher, YesWeHack Platform

May 2024 - Present

- Assessed GovTech GBBP12/13, Land Transport Authority, and 13+ other programs for SSRF, CSRF, SQL/NoSQL injection, authentication bypass, and authorization weaknesses.
- Used Burp Suite, Wireshark, Python, Bash, and custom fuzzing scripts to automate endpoint discovery, payload delivery, and evidence collection for responsible disclosure reports.

Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

SELECTED PROJECTS AND COMPETITIONS

AsyncDDGS - Open-Source Search Client

2024 - Present

- Maintained an asyncio-first, aiohttp-based DuckDuckGo search client on PyPI with pytest CI/CD and adoption across AI tooling, Discord bots, and search-enabled projects.

Arcane Security Tooling Family

2026

- Built Rust-forward security and red-team tooling experiments spanning Arcane, Arcane-PP, Arcane-GLM, and Arcane-OCR for automation, OCR, and adversarial workflows.

DSTA BRINHACK 2025 CTF

2025

- Placed in the top quartile among 400+ teams in a first CTF, applying web, crypto, reverse engineering, and systems security problem-solving under time pressure.

AgeWellLah.AI - NUS HealthHack 2025

2025

- Engineered an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, Flask, OAuth2-secured patient profiles, and PDPA-aware design choices.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Burp Suite; Wireshark; SSRF; CSRF; SQL Injection; NoSQL Injection; Authentication Bypass; OAuth2/JWT; OWASP Top 10; Python; Bash; Rust; Fuzzing; Reverse Engineering.

- Reverse Engineering Windows Executables - digital forensics and executable analysis.
- Docker for the Absolute Beginner - container fundamentals and deployment workflows.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.